

## Recommended LLM Models

After surveying current LLM APIs, the top picks are Google's **Gemini 2.5 Flash** (model ID `gemini-2.5-flash`) and OpenAI's GPT-5.6 series (e.g. **GPT-5.6 Luna** `gpt-5.6-luna` and **GPT-5.6 Terra** `gpt-5.6-terra`). These offer the best balance of Japanese fluency (including casual slang and keigo), JSON-structured output support, and cost.

- **OpenAI GPT-5.6 Luna (and Terra)** – A top-tier OpenAI model family (Luna is cost-optimized, Terra is mid-tier) with strong Japanese performance. Expert comparisons note ChatGPT/GPT models tend to produce more natural, idiomatic Japanese than competitors [41†L15-L22]. In structured-output tests, GPT models are very reliable when using function calling (“tools”) or the new “json\_schema” mode [55†L1110-L1118]. Pricing (per 1M tokens) is **\$1.00 input, \$6.00 output** for `gpt-5.6-luna` [67†L1023-L1031], and **\$2.50/\$15.00** for `gpt-5.6-terra` [67†L929-L937]. Both support Node.js via OpenAI's official SDK (the `openai` npm package) and have generous context windows (up to ~1M tokens) and ChatGPT-style function-calling support (strict JSON output via function calls). OpenAI's tiered rate limits (Tier1:500k tokens/minute, up to Tier5) cover our low-volume use [67†L1071-L1079].
- **Google Gemini 2.5 Flash** – A very low-cost Google model with good Japanese capability. Although Google's Gemini tends to favor factual accuracy, it is noted to be “warm-hearted” and culturally fluent in Japanese business contexts [51†L96-L100]. Recent surveys contrast Gemini with ChatGPT: Google's models are seen as more logically structured but ChatGPT more “natural and expressive” in Japanese [41†L15-L22]. Gemini supports structured JSON via its JSON Schema mode (Google AI's Interactions API) [53†L225-L233] and has a JavaScript/TypeScript SDK (`@google/genai`) that natively handles schemas. Pricing is extremely cheap: **\$0.30 per 1M input tokens and \$2.50 per 1M output tokens** [22†L868-L872] for `gemini-2.5-flash`. Google also offers a generous free tier (Gemini Flash models have ~1,500 requests/day or ~1 M tokens/day free) [68†L91-L100]. (Node.js usage is straightforward with Google's GenAI SDK.) The main trade-off is slightly lower nuance than GPT-5 models in casual/slang – but Gemini excels at correct formality and will reliably output valid JSON via its schema mode.

## Quality and Registers

- **Casual/slang Japanese:** GPT-based models (ChatGPT/GPT-4/5.6) are reported to generate very natural casual Japanese, matching human informal style in benchmarks [41†L15-L22]. In examples, GPT output included colloquial phrases (e.g. 「マジで」) and appropriate slang comfortably. Gemini (2.5/3.x) is generally competent in Japanese but has been noted to sound more formal or “businesslike” in style.
- **Business keigo:** Claude has been praised for impeccable honorifics [5†L255-L263], but GPT-4/5 also handle keigo well in practice. In one large Japanese survey, Claude 3 Opus scored highest on Japanese-language exam items, with GPT-4 close behind [31†L92-L100]. The AI-Market test found

all models could use keigo correctly, but GPT tended toward “model student” formality, Claude offered advanced logic in refusals, and Gemini was praised for understanding Japanese business nuances [51†L96-L100] . In short, GPT-5.6 and Gemini both handle keigo, but GPT may require more careful prompting to add warmth (Gemini by default uses polite/formal tone). All recommended models support fine-control via prompt and function/schema enforcement to ensure the correct register.

## Structured Output Support

Both OpenAI and Google fully support structured JSON output:

- **OpenAI GPT (GPT-5.6, etc.):** Use the **function-calling** (tool) API to enforce a JSON schema. OpenAI’s Structured Output guide confirms that function calling (or the new `json_schema` `response_format`) produces valid JSON and guarantees schema adherence [55†L1110-L1118] . In practice, GPT-5.6 (like GPT-4) obeys function calls with high fidelity. (Older GPT-4 variants also support this.)
- **Google Gemini:** The Gemini Interactions API explicitly supports JSON Schemas [53†L225-L233] . You supply a schema (or Pydantic/Zod schema), and the model outputs valid JSON. This is robust for data extraction or any tool-like output.

In our case, both OpenAI and Gemini SDKs let you *require* a JSON schema. The output will then be machine-parseable. Thus either provider can deliver reliable JSON output (with function-calling “strict” enforcement). Note: truly **strict** schema enforcement is best via function calling; plain “response\_format” mode in GPT may be less reliable in edge cases [55†L1110-L1118] , so using function calls with `strict=true` is recommended for safety.

## Pricing (per 1M tokens) [67†L1023-L1031] [22†L868-L872]

- **gpt-5.6-luna:** \$1.00 (input), \$6.00 (output) – *Extremely cost-effective*.
- **gpt-5.6-terra:** \$2.50 (input), \$15.00 (output) – higher quality, still far below Claude Sonnet’s \$2/\$10 (input/output).
- **gemini-2.5-flash:** \$0.30 (input), \$2.50 (output) – *very cheap*, roughly 1/6th the output cost of GPT-5.6 Luna.

Official pricing pages: OpenAI’s model comparison shows these rates for GPT-5.6 Luna and Terra [67†L929-L937] [67†L1023-L1031] . Google’s pricing page lists Gemini 2.5 Flash at \$0.30/\$2.50 [22†L868-L872] .

## SDK, Limits, and Gotchas

- **SDKs:** Both OpenAI and Google offer Node.js/TypeScript SDKs. OpenAI has an official [ `openai` npm package], and Google provides the GenAI SDK ( `@google/genai` ) with support for JSON schemas [53†L299-L307] . These make integrating LLM calls and parsing JSON straightforward in a TypeScript app.

- **Rate Limits:** For our low-volume (20–200 calls/day) use, limits are not a big issue. OpenAI GPT-5.6 Luna/Terra have high tiered per-minute quotas (Tier1 starts at 0.5M tokens/min 【67†L1071-L1079】 ) far above our needs. Google Gemini Flash has a free tier (1,500 requests/day, ~1M tokens/day) and a paid tier with similar or higher limits. So both run reliably at light usage.
- **Regional Availability:** OpenAI’s API is global (some users in SA use Azure/OpenAI too). Google’s Gemini API is also global via Google Cloud. There are no known geo-restrictions that would affect a South African developer for either service. (Claude/Anthropic is generally USA-only, but we’re not using Claude here.)
- **Output Consistency:** While no LLM is 100% perfect, using function calls/schema enforces structure. Both GPT and Gemini have demonstrated good “schema compliance” with function-calling. In practice, GPT-5.6 with `{"name": "function", "arguments": schema}` mode yields nearly guaranteed valid JSON. Google’s structured output mode similarly locks to schema 【53†L225-L233】 . So neither is “best effort” free-form; both support *forced* structured output.

## Recommendation

**GPT-5.6 Luna** is the clear best pick if budget allows: it matches or exceeds Claude’s Japanese fluency (ChatGPT consistently tops Japanese benchmarks) and costs only ~\$6 per million output tokens 【67†L1023-L1031】 . Its function-calling is reliable, and it supports casual slang and keigo well.

For tighter budgets, **Gemini 2.5 Flash** is a strong runner-up: at ~\$0.30/\$2.50 per million 【22†L868-L872】 it is far cheaper than Claude Sonnet and still produces very good Japanese. (It tends toward a slightly more formal tone by default, but prompt tuning can inject casual style.)

Finally, **GPT-5.6 Terra** provides a quality-focused option (ideal if a bit higher cost is acceptable). Overall we recommend GPT-5.6 Luna (or Terra) as the best all-around, with Gemini 2.5 Flash as the low-cost alternative.

**Sources:** Official OpenAI and Google pricing and docs 【67†L929-L937】 【67†L1023-L1031】 【22†L868-L872】 【53†L225-L233】 , plus recent Japanese LLM evaluations and expert comments 【41†L15-L22】 【51†L96-L100】 【5†L255-L263】 .

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